

Freelance Projects

1. Automated Blood Cell Counting

- *What:* I automated the task of counting different types of blood cells under a microscope (designed for a pathology).
- *Why:* A lab-technician has to count hundreds of cells one by one. This includes human errors and mistakes.
- *How:* I have used CNN for object detection task. The model scans the image, find every cell, and puts a little box around it to identify if it's a red cell, white cell, or platelet.
Also, different labs use different dyes (some are more purple, some more pink). I added a 'color normalization' step so that the model doesn't get confused by different stains. This way, my CNN models identify cells (even overlapping cell) easily.
- *Dataset:* The pathology dataset, has thousands of images of blood smears with different types of cells labeled.
- *Output:* A full report showing the count and percentage of each cell type.

2. Mapping Tumors in Tissue Scans

- *What:* I worked on a tool to help doctors see exactly where a tumor starts and ends in a high-resolution tissue scan.
- *Why:* Doctors have to manually trace the edges of a tumor. If they miss a tiny bit, it's dangerous.
- *How:* Used a U-Net CNN architecture. It first understands what the cancer looks like, and then it draws precise map of it. I used 'skip connection' technique, to let the model remember the sharp edges of the original image while making prediction. This CNN model classifies cancer and draws the boundary, useful during surgery. Every single pixel in the image is given a score between 0 (healthy) and 1 (cancer). By grouping all the high score pixels together, the model creates a Binary Mask (a black-and-white shape) that perfectly outlines the tumor.
- *Dataset:* The pathology data, containing WSI of lymph nodes from breast cancer patients.
- *Output:* A color-coded map (a heatmap) that overlays directly on the doctor's screen to show the tumor boundaries.

3. Measuring Cancer Growth (Mitosis)

- *What:* I built a tool to measure how fast a cancer is growing by finding cells that are currently dividing.
- *Why:* To know how aggressive a cancer is, doctors look for Mitotic Figures (cells splitting in two). These are tiny and hard to find.
- *How:* I trained a CNN to be a 'Specialized Searcher'. It is trained to focus on the specific shape of a cell when its DNA is pulling apart. Since these cells are so rare, I used 'Data Augmentation'. I flipped and rotated the few examples I had so the model could recognize them from any angle.
- *Dataset:* Pathology dataset, manually annotated image data.
- *Output:* A Mitotic count (e.g., 10 dividing cells per square millimeter) which gives the cancer its grade (severity).

Self-Projects

1. Spam Detection

- *What:* I have built a system to filter out junk messages without slowing down the user's app.
- *Why:* Most filters look for specific bad words. Spammers know this and just change the spelling. I needed a system that can understand the pattern of a spam message, not just the words.
- *How:* Converted words into vector so the computer could see which words are similar. I used 1D-CNN with 3-word limit (window size =3). I used 'max pooling' layer to grab the most suspicious phrase from any part of the message. This helps the model to recognize phrases like "win cash now" as a single red flag.
- *Dataset:* Used the UCI SMS Collection dataset, with 5000 text messages labeled as 'ham' or 'spam'.
- *Output:* A simple 'safe' or 'spam' label with a confidence score.

2. Neural Disorder Detection via Brain Imaging

- *What:* I built a system that listens to the brain's electrical rhythm to predict or detect disorders before they become visible to the human eye.
- *Why:* Brain signals (EEG) are messy. A doctor has to look at wavy lines to find one 2-second glitch that indicates a seizure or a disorder.

- *How:* I used a 1D- CNN to find the spikes and shape of a brain wave (local patterns) and an RNN (specifically an LSTM) to remember the history of those waves (the sequence). This is crucial because a wave might look normal on its own, but if it follows a specific sequence of other waves, it indicates a disorder.
I used Fast Fourier Transform (FFT) to convert wavy lines into frequency bands (alpha, beta, and delta waves).
- *Dataset:* CHB-MIT EEG Database, which has digital EEG signals, comprising hundreds of hours of continuous recordings from multiple patients.
- *Output:* Labeled as 'Interictal' (normal brain activity) and 'Ictal' (active seizure or disorder state).

3. Autism Identification via Brain Imaging

- *What:* I developed a system to identify structural and functional brain anomalies in children using deep learning.
- *Why:* Reading fMRI or MRI scans is extremely difficult because a child's brain is constantly changing. Distinguishing between a developing brain and a disordered brain requires extreme precision.
- *How:* Used Transfer Learning with a 3D-CNN. This model is already trained on millions of images, and fine-tuned its final layer to recognize the subtle frequency anomalies in a child's brain structure.
- *Dataset:* I used the Autism Brain Imaging Data Exchange (ABIDE) dataset consists of thousands of brain scans report from children with and without autism.
- *Output:* A Saliency Map (a brain heatmap), highlighting the exact parts of the brain showing neurological disorders.

4. Autism Identification through Digital Touch Dynamics

- *What:* I developed a way to screen children for Autism simply by getting simple tasks done in mobile/tablets. The model looks at how they move their fingers.
- *Why:* Traditional autism screening is subjective and happens too late (usually age 4 or 5). I wanted a way to find digital red flags as early as age 2, using a tool kid already use, like phone or tabs.
- *How:* Used a CNN to analyze the movement. I looked for Kinematic Markers, specifically Jerk (how shaky a movement is). I treated the

touch data as a 1D signal. The CNN slides across the pressure and speed data to find specific shivers or stiffness in the finger movements that are invisible to the human eye. I focused on Fine motor Signatures to capture the micro-movements.

- *Dataset:* <https://gitlab.citic.udc.es/lia2-publico/aitouch-code> consists of data tracks of every micro-movement on the screen.
- *Output:* A Risk Profile Report. It gives a probability score for autism and highlights specific motor-coordination areas where the child might need extra support or therapy.

5. Multi-Modal Behavioral Disorder Detection

- *What:* I developed an AI system that mimics how a doctor evaluates a child, by watching their facial expressions while simultaneously observing how they physically interact with objects. It's a multi-modal approach to identify developmental atypicality.
- *Why:* Autism and other neurological disorders are complex. If you only look at video, you might miss physical tremors. If you only look at touch data, you miss emotional cues. I wanted to build a brain that could process both at the same time for a more accurate screening.
- *How:* I built a dual network:
 1. The CNN framework: Uses a ResNet-18 backbone to extract facial features (like a lack of eye contact or flat affect).
 2. The RNN framework: Uses an LSTM to watch the sequence of touch movements. It looks for jerk and pressure patterns. This is where we detect motor-coordination issues.

I used a Feature Fusion technique. Instead of having two separate models, the AI merges the visual and kinetic data into a single 'Feature Vector'. This allows the model to learn, for example, that a specific hand tremor (Kinetic) occurring during a specific moment is a risk marker.

- *Dataset:* I used high-frequency kinematic data (AITouch and MDPI Dataset) and paired it with visual frames (Self-Stimulatory Behavior Dataset).
- *Output:* A Comprehensive Behavioral Score that flags whether a child shows 'Typical' or 'Atypical' developmental markers.